

YUPU YAO

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EDUCATION

University of Electronic Science and Technology of China & University of Glasgow

Dual B.Eng., Communication Engineering (Joint Program)

Sept 2022 - Expected June 2026

PUBLICATIONS AND MANUSCRIPTS

[Published]

Yao, Y. (2024). Generalize neural network through smooth hypothesis function. *In The Second Tiny Papers Track at International Conference on Learning Representations (ICLR 2024)*.

[Under Review]

Yao, Y., Yang, B. (2025). GeoPE: A Unified Geometric Positional Embedding for Structured Tensors. *Submitted to International Conference on Learning Representations (ICLR 2026)*.

[Preprint]

Yao, Y., Deng, S., Cao, Z., Zhang, H., & Deng, L. (2024). APLA: Additional perturbation for latent noise with adversarial training enables consistency. *arXiv preprint arXiv:2308.12605*. (Cited by 15, Google Scholar)

INTERNSHIP EXPERIENCE

Infinigence AI

Inference Framework Developer Intern

Aug 2025 - Present

Shanghai, China

- Led the development of an SGLang-based and vllm-based heterogeneous inference framework, architecting an efficient GPU-FPGA scheduler to resolve distributed Attention and MoE bottlenecks via computation-communication overlap.
- Implemented Prefill/Decode phase separation to mitigate mutual interference between processing stages and enhance overall throughput.
- Successfully deployed and optimized large-scale models (DeepSeek 671B, Qwen 235B) supporting FP8/INT4 precision on distributed 8xH20 and 8xA100 clusters.

RESEARCH EXPERIENCE

Deconstructing Attention Mechanisms via Implicit Hilbert Ladder

Feb 2024 - June 2024

- Proposed and implemented the Implicit Hilbert Ladder (IHL) framework, modeling Multi-Head Self-Attention (MSA) from a Reproducing Kernel Hilbert Space (RKHS) perspective to provide a new theoretical understanding of Transformers.
- Developed a novel mechanism, Hilbertian Attention Cascade (HAC), replacing self-attention with kernel mappings in RKHS. Achieved a >5% accuracy increase in early-stage training on CIFAR-10, with 16% fewer parameters and a 12.3% reduction in GFLOPs.
- Validated the theoretical advantages of IHL via Rademacher complexity generalization bounds, Neural Tangent Kernel (NTK) convergence proofs, and gradient flow analysis, establishing a theoretically-grounded attention alternative

Resource Allocation for Semantic-Aware Communication Networks

Aug 2022 - Feb 2024

- Deployed and trained a Deep Q-Network (DQN)-based intelligent resource allocation algorithm, improving average user Quality of Experience (QoE) by 23.5% and system convergence speed by 18% in simulated wireless networks.
- Proposed a distributed two-tier control strategy integrating local DQNs with a global coordination module, reducing training communication overhead by 28.6% while maintaining near-optimal global performance.
- Effectively addressed the joint task of beamforming, power control, and semantic resource allocation, demonstrating a 19% improvement in average policy robustness under multi-task, high-dimensional state spaces.

VQ-GAN based Semantic Communication System

Dec 2022 - Mar 2023

- Proposed and implemented a novel end-to-end semantic communication architecture, introducing a VQ-GAN encoding module to replace traditional source-channel separation and achieve joint optimization of compression and representation.
- Improved semantic reconstruction accuracy by 21.3% and bandwidth efficiency by 35.7% compared to conventional systems.

ENGINEERING PROJECT

Development of a Multifunctional Autonomous Navigation Drone

Mar 2025 - June 2025

- Designed and deployed a fully autonomous UAV system by integrating a PX4 flight controller with an OrangePi edge device, achieving >98% flight stability during automated missions.
- Integrated and optimized a multi-module algorithm stack (VINS-MONO, Metric3D, YOLOv8n, EGO-Planner), reducing end-to-end system response latency by 30%.
- Architected the system on ROS 2 (Humble), optimizing communication between control nodes and sensors and reducing CPU occupation to 60% of the original baseline.

Construction of a 6G Communication Simulation Platform

Jan 2024 - Dec 2024

- Led the design and integration of pilot-aided channel estimation algorithms for a multi-university 6G simulation platform.
- Established a unified API standard to streamline system integration, shortening debugging cycles by over 30%.
- Developed an interactive front-end interface for visualizing channel estimation results and enabling real-time parameter tuning.

AWARDS AND SCHOLARSHIPS

Provincial Gold Medal, The 9th China "Internet+" College Students' Innovation Competition

Aug 2023

Developed and deployed advanced remote sensing algorithms for applications in agriculture, environmental monitoring, disaster prediction, and defense.

TECHNICAL SKILLS

- **ML/DL Frameworks:** Pytorch, Tensorflow, JAX, Triton
- **Inference Frameworks:** SGLang, vllm
- **Technical Domains:** Computer Vision (GAN, Diffusion, Flow), Reinforcement Learning (DQN, PPO), NLP (Transformer), Large Models (RLHF, RAG)
- **Learning Paradigms:** Supervised Learning, Deep Learning, Reinforcement Learning, Few-shot Learning, Transfer Learning